

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 2

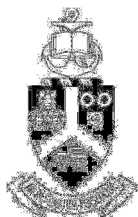
Kopiereg voorbehou

Semester Test 2

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Module EBN111
Elektrisiteit en Elektronika
8 Mei 2015

Module EBN111
Electricity and Electronics
8 May 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: *Exam information:*

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		15	

BELANGRIK- IMPORTANT

1. Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
2. Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
3. Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
4. Antwoorde moet eenhede insluit!
Answers must include units!
5. **Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 A Chatterjee

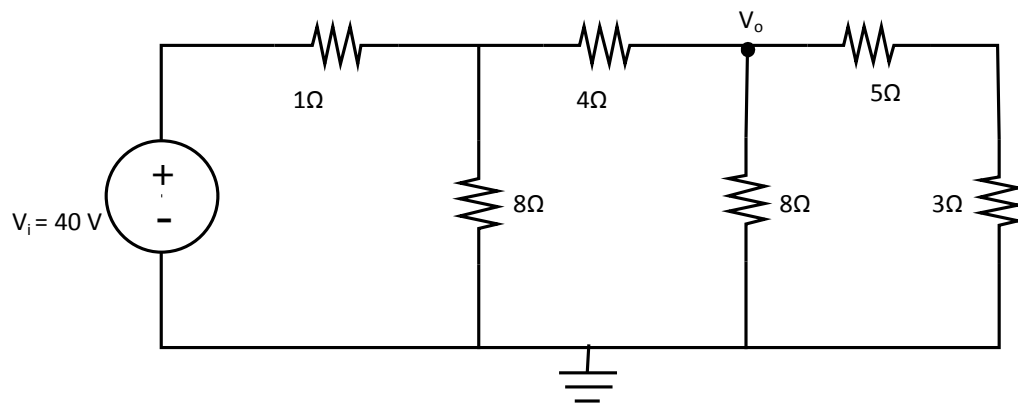
VRAAG/QUESTION 1 [23]

Vraag 1.1 [3]

- a) Aanvaar $V_o = 1\text{ V}$ en gebruik lineariteit om V_o te bereken.
b) Bepaal V_o as V_i verander word na 15 V .

Question 1.1 [3]

- a) Assume $V_o = 1\text{ V}$ and use linearity to find the actual value of V_o .
b) If V_i is changed to 15 V , find V_o .

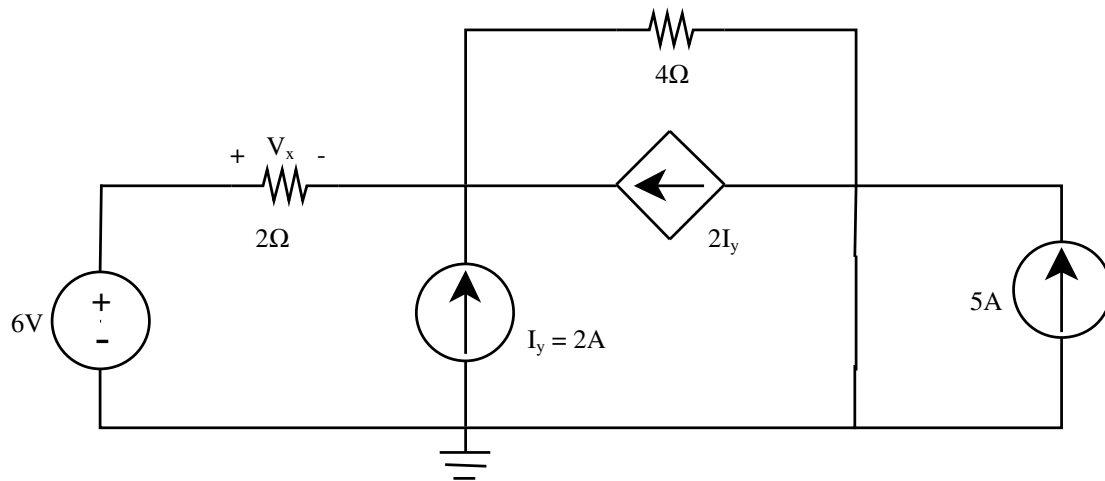


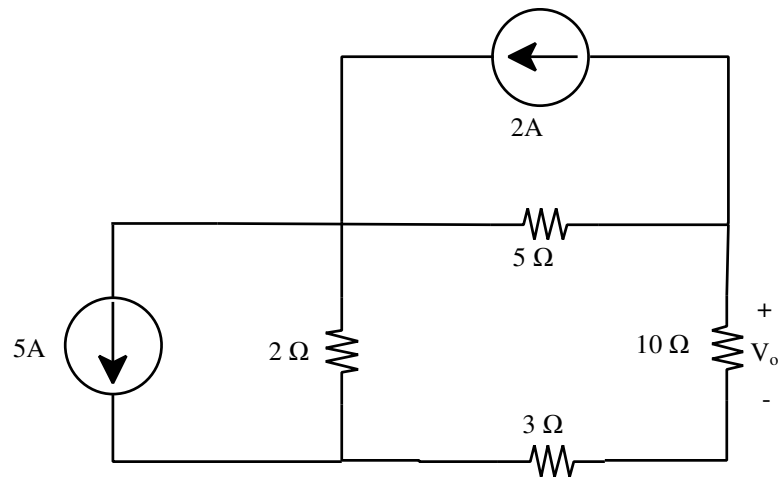
Vraag 1.2 [4]

- a) Bepaal V_x slegs agv die 6V bron.
- b) Bepaal V_x slegs agv die 2A bron.
- c) Bepaal V_x agv al die bronne tesaam.

Question 1.2 [4]

- a) Find V_x due to the 6V source only.
- b) Find V_x due to the 2A source only.
- c) Find V_x due to all the sources together.



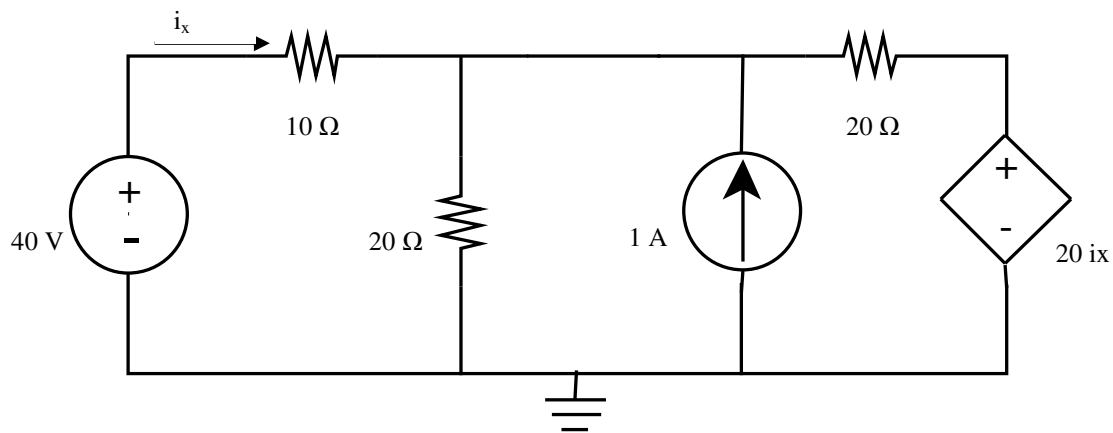
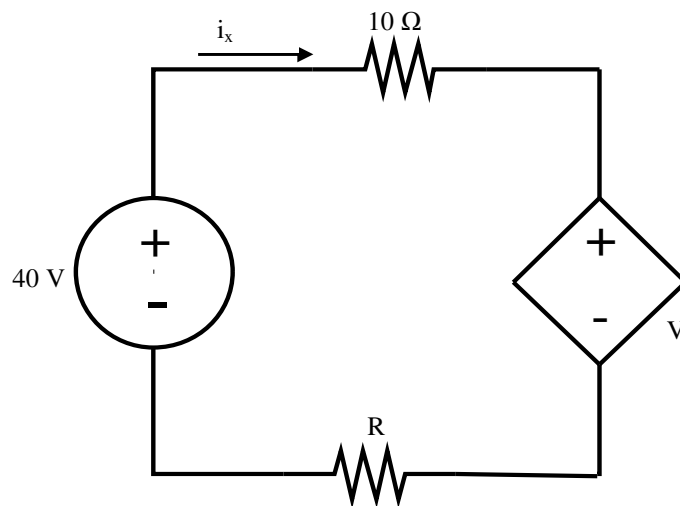
Vraag 1.3 [2]Bereken V_o mbv brontransformatie.**Question 1.3 [2]**Find V_o by using source transformation.

Vraag 1.4 [3]

Gebruik brontransformatie om die stroombaan in fig. 1 te transformeer sodat dit lyk soos fig. 2. Bepaal R , V en i_x .
(wenk: V moet 'n funksie van i_x wees).

Question 1.4 [3]

Use source transformation to transform the circuit in fig. 1, to the circuit shown in fig. 2. Find R , V , and i_x .
(Hint: V should be a function of i_x).

**Figure 1****Figure 2**

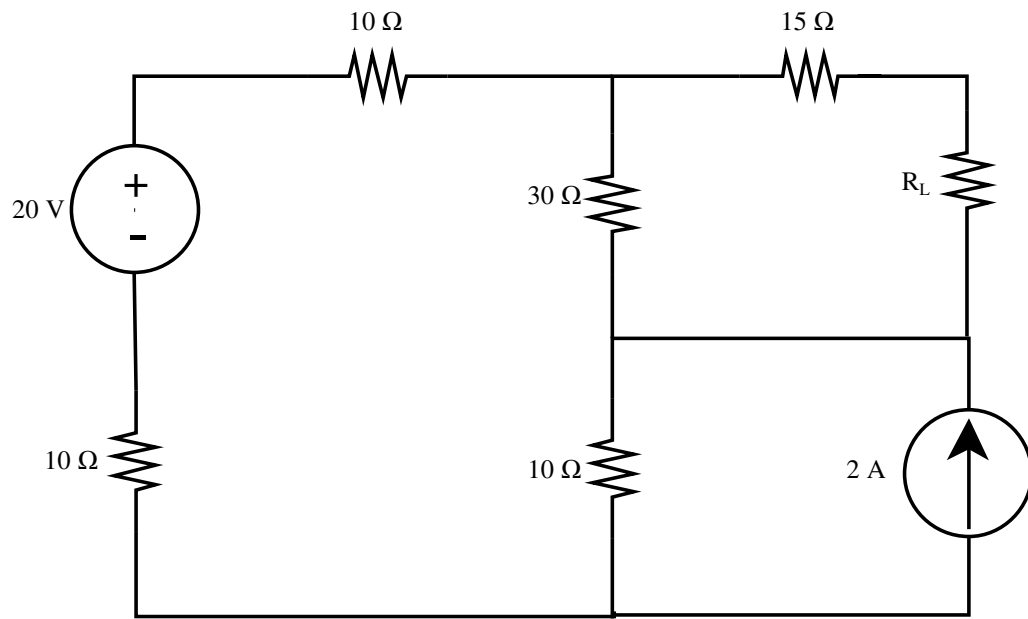
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Vraag 1.5 [4]

- a) Bepaal die Thevenin-weerstand soos gesien deur die lasweerstand, R_L .
b) Bepaal die Thevenin-spanning soos gesien deur R_L .

Question 1.5 [4]

- a) Find the Thevenin resistance as seen by the load resistance, R_L .
b) Find the Thevenin voltage as seen by R_L .



Vraag 1.6 [1]

Indien 'n stroombaan 'n Thevenin ekwivalente spanning $V_{th} = 50V$ het, en 'n Norton ekwivalente stroom $I_N = 5A$ het; bereken hoeveel stroom deur 'n 50Ω lasweerstand sal vloei as dit aan die baan gekoppel sou word.

Question 1.6 [1]

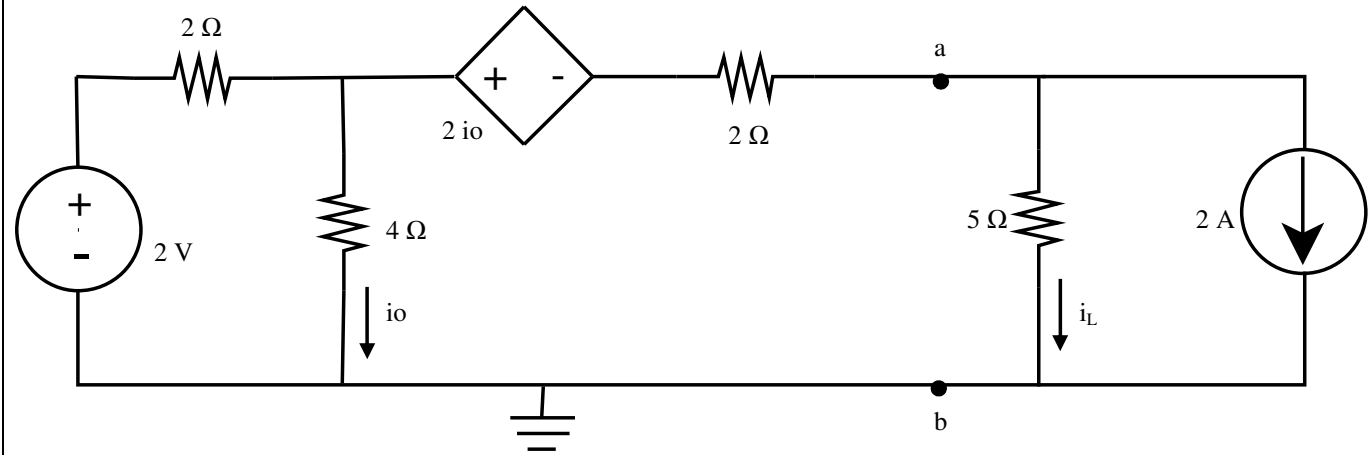
If a circuit has a Thevenin equivalent voltage $V_{th} = 50V$, and a Norton equivalent current $I_N = 5A$; calculate how much current a 50Ω load, connected to this circuit, would draw.

Vraag 1.7 [6]

- a) Bepaal die Norton-weerstand, soos gesien links van terminale a-b.
- b) Bepaal die Norton-stroom, soos gesien links van terminale a-b.
- c) Aanvaar $R_N = 10\ \Omega$ en $I_N = 5\text{ A}$. Bepaal die lasstroom I_L .

Question 1.7 [6]

- a) Find the Norton resistance, as seen to the left of terminals a-b.
- b) Find the Norton current, as seen to the left of the terminals a-b.
- c) Assume $R_N = 10\ \Omega$ and $I_N = 5\text{ A}$. Find the load current I_L .



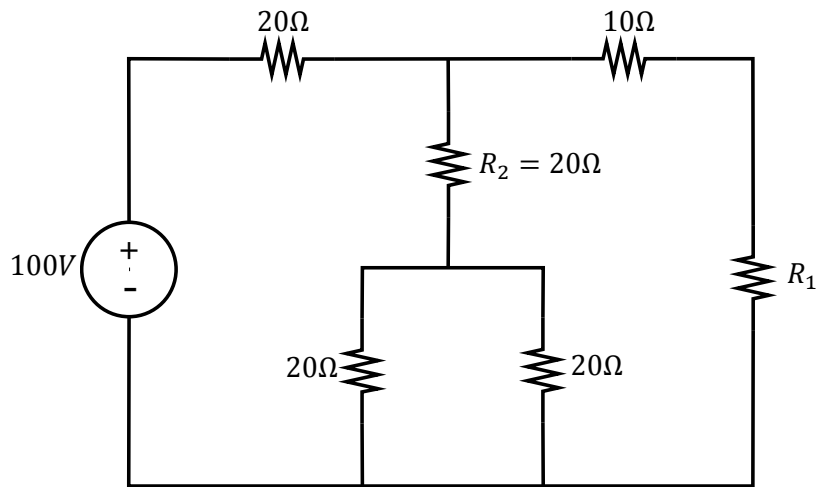
VRAAG/QUESTION 2 [22]

Vraag 2.1 [4]

- a) Bepaal weerstand R_1 sodat weerstand R_1 die maksimum drywing sal dissipeer.
- b) Bepaal weerstand R_1 sodat weerstand R_2 die maksimum drywing sal dissipeer.

Question 2.1 [4]

- a) Find the resistance R_1 such that resistor R_1 will dissipate maximum power.
- b) Find the resistance R_1 such that resistor R_2 will dissipate maximum power.

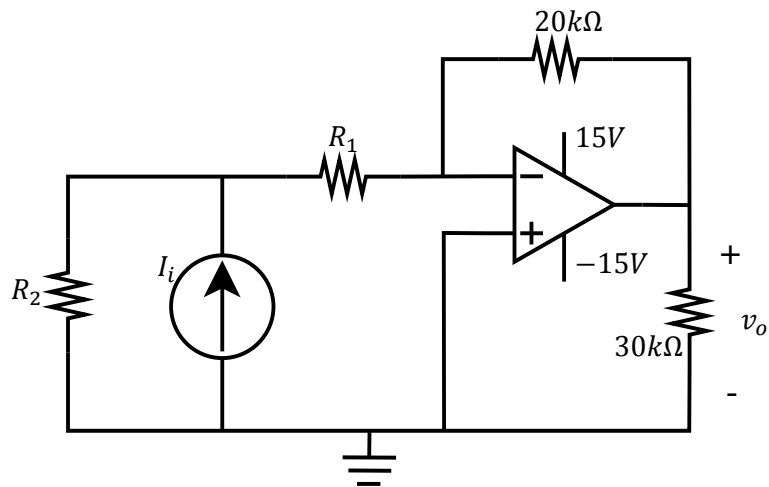


Vraag 2.2 [4]

- a) Bereken v_o as $I_i = 0.2mA$ en $R_1 = R_2 = 10\text{ k}\Omega$.
b) Bereken v_o as $I_i = 0.2mA$, $R_1 = 10\text{ k}\Omega$, en $R_2 = \infty$.

Question 2.2 [4]

- a) Determine v_o if $I_i = 0.2mA$ and $R_1 = R_2 = 10\text{ k}\Omega$.
b) Determine v_o if $I_i = 0.2mA$, $R_1 = 10\text{ k}\Omega$, and $R_2 = \infty$.

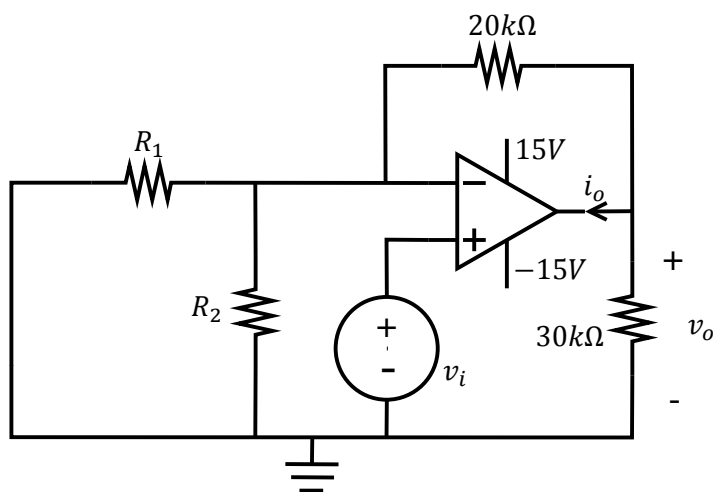


Vraag 2.3 [4]

- a) Bereken v_o as $v_i = 2V$ en $R_1 = R_2 = 20\text{ k}\Omega$.
b) Bereken i_o (in mA) as $v_i = 2V$, $R_1 = 20\text{ k}\Omega$, en $R_2 = \infty$.

Question 2.3 [4]

- a) Determine v_o if $v_i = 2V$ and $R_1 = R_2 = 20\text{ k}\Omega$.
b) Determine i_o (in mA) if $v_i = 2V$, $R_1 = 20\text{ k}\Omega$, and $R_2 = \infty$.

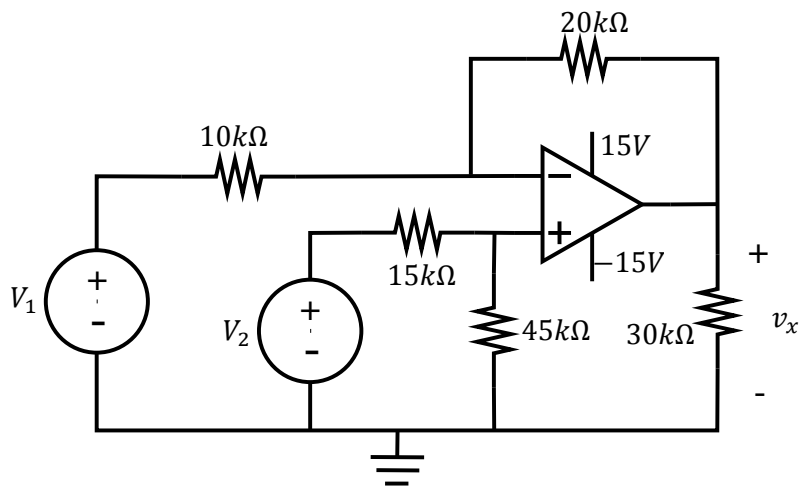


Vraag 2.4 [4]

- a) Bereken v_x as $V_1 = 3V$ en $V_2 = -2V$.
b) Wat is die minimum waarde wat V_2 kan wees voordat die OV in versadiging ingaan (aanvaar $V_1 = 3V$).

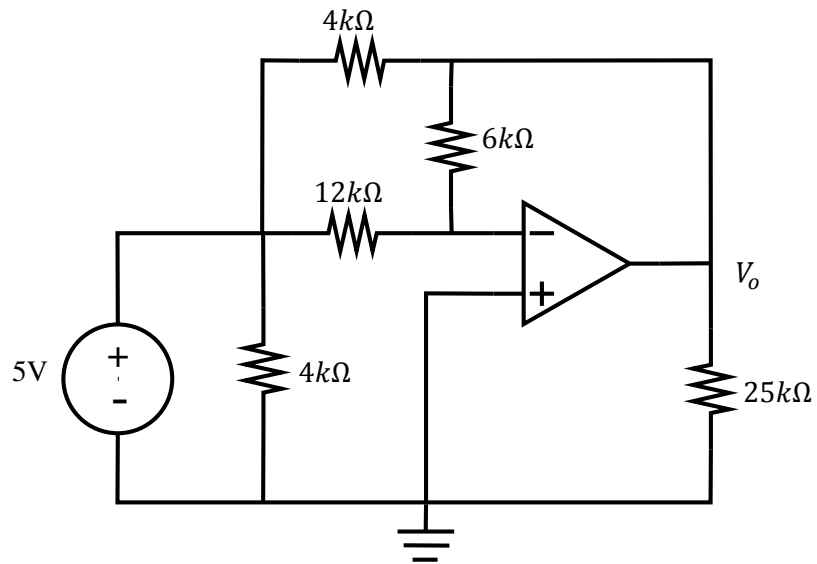
Question 2.4 [4]

- a) Determine v_x if $V_1 = 3V$ en $V_2 = -2V$.
b) What is the minimum value V_2 can be before the op-amp goes into saturation (assume $V_1 = 3V$).



Vraag 2.5 [2]
Bereken V_o .

Question 2.5 [2]
Determine V_o .

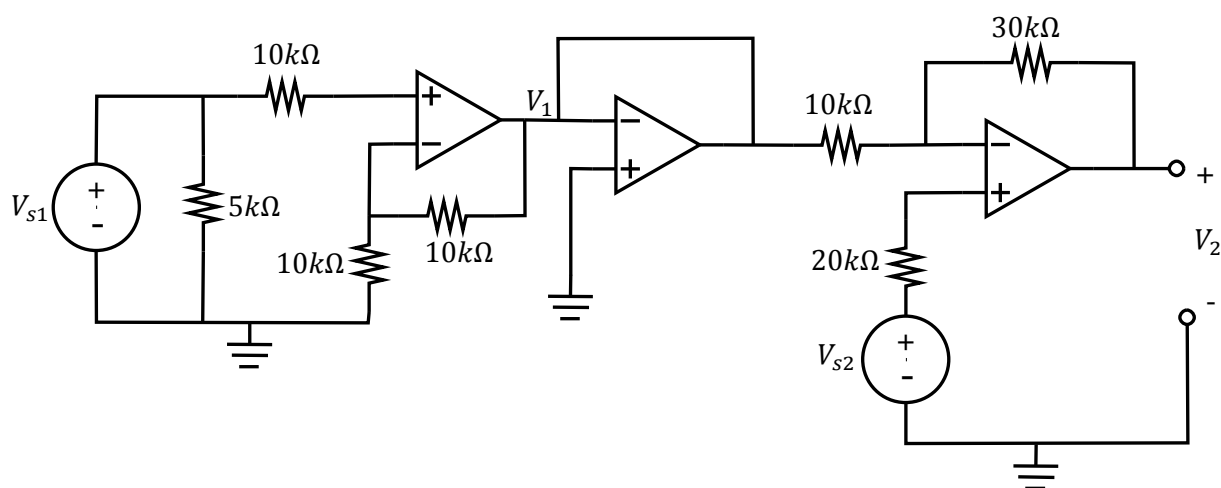


Vraag 2.6 [4]

- a) Bereken V_1 as $V_{s1} = 1mA$.
b) Bereken V_2 as $V_1 = 2V$ en $V_{s2} = 3V$.

Question 2.6 [4]

- a) Determine V_1 if $V_{s1} = 1mA$.
b) Determine V_2 if $V_1 = 2V$ and $V_{s2} = 3V$.



EBN111 SEMESTER TOETS/TEST 2 (8 MEI/MAY 2015)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language:	Afr		Eng				

Q1		Q2		Tot:	/45
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VRAAG 1 – QUESTION 1 (23)			
1.1	$a) V_o =$	$b) V_o =$	
1.2	$a) V_x =$	$b) V_x =$	$c) V_x =$
1.3	$V_o =$		
1.4	$R =$	$V =$	$i_x =$
1.5	$R_{TH} =$	$V_{TH} =$	
1.6	$I_L =$		
1.7	$a) R_N =$	$b) I_N =$	$c) I_L =$

VRAAG 2 – QUESTION 2 (22)		
2.1	$a) R_I =$	$b) R_I =$
2.2	$a) v_o =$	$b) v_o =$
2.3	$a) v_o =$	$b) i_o =$ mA
2.4	$a) v_x =$	$b) V_{2min} =$
2.5	$V_o =$	
2.6	$a) V_I =$	$b) V_2 =$